

Education			
Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	7.25/10
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	91.2%
2017 – 2020	High School (X)	SJN International School, Coimbatore	92.6%

Experience

Google Summer of Code 2025 Contributor | P4 Language Consortium, Stanford, CA (May 2025 – Sept 2025)

SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4 P4, Python, P4Runtime, Docker, C++, Intel Tofino

- Switch-Native Architecture:** Architected a framework to compile ML decision trees directly into P4 data planes, implementing Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs (SIDs) to resolve hardware ASIC constraints.
- Validation & Optimization:** Engineered a Jinja2-based P4Runtime testing pipeline (Tofino/Mininet) improving deployment reliability by 30%, and optimized SID recirculation to achieve 10 Gbps line-rate throughput with 20% lower inference latency.

Open Source Contributions

- WasmEdge/WasmEdge:** #4470 fix(wasi): resolve 100% CPU usage on macOS due to clock mismatch; #4466 feat(docker): add standalone alpine-base image build pipeline; #4440 Upgrade Alpine base to 3.23 and restore static LLD support.
- facebook/bpfilter:** #507 build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c.

Key Projects

Cutable | Go, Next.js, React, TypeScript, PostgreSQL, Neon, OpenRouter, E2B, WebSockets, Docker, GitHub Actions, Nomad, Traefik, WireGuard, Google OAuth, JWT (Project Link)

Objective: Turn natural-language and visual requirements into editable, previewable React applications.

Approach: Built a Go agent-orchestration backend around OpenRouter tool calls and isolated E2B sandboxes; streamed planning, code edits, build progress, and previews to a Next.js workspace over WebSockets with persistent project recovery.

Impact: Delivers multimodal generation, BYOK provider access, account-scoped project history, and health-gated immutable deployments through a self-hosted Nomad platform.

DMIE | Next.js, TypeScript, Python, FastAPI, Vector DB, LLM (Project Link)

Objective: Determine whether a startup idea already exists by comparing it against more than 500,000 company records.

Approach: Built a scheduled scraper that keeps the startup dataset current; combined LLM query with vector search and backend filters.

Impact: Raised USD 10,000 and serves more than 100 users worldwide.

Slix | React, Go, Azure OpenAI, REST API (Project Link)

Objective: Combine movie streaming, reviews, and machine-generated sentiment summaries in one full-stack application.

Approach: Built the frontend in React and the service layer in Go; added Azure OpenAI sentiment analysis for curator reviews.

Impact: Gives viewers a concise representation of a movie's tone before watching.

Myprod | Go, Nomad, Traefik, WireGuard, Docker, Netlify DNS, SOPS (Project Link)

Objective: Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.

Approach: Built a Go CLI and operator dashboard around Nomad scheduling, Traefik ingress, and WireGuard networking; added guarded node lifecycle controls, application registration, live resource metrics, and idempotent Netlify DNS management.

Impact: Provides one control surface for deploying applications and managing heterogeneous free-tier and temporary compute capacity.

Technical Skills

- Language:** Python, C, C++, JavaScript, TypeScript, Go, P4
- Libraries/Frameworks:** NodeJS, Numpy, Matplotlib, eBPF, gRPC, Protobufs, Next.js, Scapy, MLIR, LLVM, Sklearn, React, Web Services
- Tools/Platforms:** Docker, CI/CD, VS Code, UTM, GitHub, GitLab, DaVinci Resolve, Linux (x86_64, ARM64 & AMD64)
- Databases:** SQL, MongoDB, GraphQL, Redis, PostgreSQL, MySQL

Positions of Leadership

- Technical Writer, Cloud Native Community Group, New Delhi (May 2024 – Dec 2024)**
 - Led a team of 6+ content writers and 4 designers to deliver documentation and event management support.
 - Partnered with marketing to revamp social media presence across LinkedIn, Instagram, and X, achieving a 60% bump in engagement.
- Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR (Nov 2023 – Nov 2025)**
 - Co-led 10+ tech events & anchored the GDSC WOW flagship event at IIIT Delhi, with the 31+ crew members.
 - Engaged a live audience of 800+ at WOW and 500+ developers overall, boosting participation by 30% and visibility by 40%.

Publications

- R. Bharadwaj, S. Jha, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," 2026 International Conference on Computing, Sciences and Communications (ICCS), Ghaziabad, India, Feb. 2026. IEEE Xplore ↗ DOI: 10.1109/ICCS67078.2026.11468597

Certifications & Honors

Certifications	• Open Science 101 – NASA • Data Science for Engineers – NPTEL • Enhancing Soft Skills & Personality – NPTEL • IIRS Satellite Remote Sensing – ISRO
Honors & Awards	• 2x AKTU State Men Singles TT Gold (2023, 2024) • International Robotics Olympiad Bronze, Thailand (2019) • Merit Certification at Carnic Model United Nation (2021) • NCC Cadet, National Cadet Corps (2019)